

The coefficients of the binary number are obtained from the integers as follows:

	integer	fraction	coefficient
$0.6875 \times 2 =$	1	0.3750	$a_1 = 1$
$0.3750 \times 2 =$	0	0.7500	$a_2 = 0$
$0.7500 \times 2 =$	1	0.5000	$a_3 = 1$
$0.5000 \times 2 =$	1	0.0000	$a_4 = 1$

answer: $(0.6875)_{10} = (0.a_1 a_2 a_3 a_4)_2 = (0.1011)_2$

To convert a decimal fraction to a number expressed in base r , a similar procedure is used. Multiplication is by r instead of 2, and the coefficients found from the integers may range in value from 0 to $r-1$ instead of 0 and 1.

Imp.

Example 1-4: Convert $(0.513)_{10}$ to octal

$$\begin{aligned} 0.513 \times 8 &= 4.104 \\ 0.104 \times 8 &= 0.832 \\ 0.832 \times 8 &= 6.656 \\ 0.656 \times 8 &= 5.248 \\ 0.248 \times 8 &= 1.984 \\ 0.984 \times 8 &= 7.872 \end{aligned}$$

The answer, to seven significant figures, is obtained from the integer part of the products: $(0.513)_{10} = (0.406517 \dots)_8$

1.4 Octal and Hexadecimal Numbers

The conversion from and to binary, octal and hexadecimal plays an important part in digital computers. Since $2^3 = 8$ and $2^4 = 16$, each octal digit corresponds to three binary digits and each hexadecimal digit corresponds to four binary digits. The conversion from binary to octal is easily accomplished by partitioning the binary number into groups of three digits each, starting from the binary point and proceeding to the left and to the right.

$$\begin{array}{ccccccc} (19 & 119 & 001 & 100 & 011 & 111 & 100 & 000 & 110)_2 & = & (26153.7406)_8 \\ \begin{array}{ccccccc} \boxed{19} & \boxed{119} & \boxed{001} & \boxed{100} & \boxed{011} & \boxed{111} & \boxed{100} & \boxed{000} & \boxed{110} \\ \hline 2 & 6 & 1 & 5 & 3 & 7 & 4 & 0 & 6 \end{array} \end{array}$$

Conversion from binary to hexadecimal is similar, except that the binary number is divided into groups of four digits:

$$\begin{array}{ccccccc} (10 & 1100 & 0110 & 1011 & 1111 & 0010)_2 & = & (2C6B.F2)_{16} \\ \begin{array}{ccccccc} \boxed{10} & \boxed{1100} & \boxed{0110} & \boxed{1011} & \boxed{1111} & \boxed{0010} \\ \hline 2 & C & 6 & B & F & 2 \end{array} \end{array}$$